7.2 Factoring a Trinomial With Algebra Tiles

Lesson Introduction

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<tr><td>Benchmark</td><td colspan="2">MA.912.AR.1.7 Rewrite a polynomial expression as a product of polynomials over the real-number system.</td><td>Learning Target</td><td colspan="2">• I can use algebra tiles to factor a trinomial with a leading coefficient of 1.</td></tr>
<tr><td rowspan="2">Benchmark Coherence</td><td colspan="2">Building On</td><td colspan="3">Working Toward</td></tr>
<tr><td colspan="2">MA.8.AR.1.3 Rewrite the sum of two algebraic expressions having a common monomial factor as a common factor multiplied by the sum of two algebraic expressions.</td><td colspan="3">MA.912.AR.1.8 Rewrite a polynomial expression as a product of polynomials over the real- or complex number system.</td></tr>
<tr><td>Essential Questions</td><td colspan="5">• How can algebra tiles be used to show factoring a trinomial with a leading coefficient of one?
• What is the relationship between multiplication and factoring polynomials?</td></tr>
<tr><td>Lesson Narrative</td><td colspan="5">In this lesson, students use algebra tiles to explore factoring trinomials. Students will connect multiplying binomials with algebra tiles to using that information to factor trinomial expressions. They will then spend time practicing the process of factoring trinomials with a leading coefficient of one.</td></tr>
<tr><td rowspan="3">Component Summary</td><td>7.2.1 Warm-Up (~5 min.)</td><td>Multiply binomials using algebra tiles (SE p. 363)</td><td>7.2.3 Guided Practice (10 min.)</td><td colspan="2">Practice factoring trinomials using models (SE p. 368)</td></tr>
<tr><td>7.2.2 Exploration (20 min.)</td><td>Explore factoring trinomials using algebra tiles (SE p. 363)</td><td>7.2.4 Your Turn! (5 min.)</td><td colspan="2">Independent practice factoring trinomials using models (SE p. 368)</td></tr>
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<tr><td rowspan="2">Resources</td><td>Glossary</td><td colspan="2">Materials</td><td colspan="2">Additional Preparation</td></tr>
<tr><td><u>Key Terms:</u> N/A

<u>Additional Terms</u>:
• Factoring</td><td colspan="2">• (Optional) Algebra tiles</td><td colspan="2">• N/A</td></tr>
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<tr><td rowspan="4">Teacher Tips</td><td>Common Misconceptions</td><td>Things to Consider</td></tr>
<tr><td>Students may make errors with the signs of the algebra tiles.
Students may reverse the sum and product of the trinomial factoring process.</td><td>The conceptual foundation that the lesson lays should not be rushed. Allow students to explore and ask questions.
Using algebra tiles to model is a powerful tool to use in building students’ understanding of this concept. If you are unfamiliar with using these models, be sure to watch the video for this lesson to see how it unfolds in this structure.</td></tr>
<tr><td>Literacy and Language Routines</td><td>Mathematical Thinking and Reasoning (MTR) Standard</td></tr>
<tr><td>Take Turns
7.2.2 Exploration is a collaborative exploration activity where students work together to model factoring with algebra tiles. During this activity, consider leveraging this routine to have students take turns with the tiles and steps within the process.</td><td>MA.K12.MTR.2.1: Multiple Representations
7.2.2 Exploration has students working with algebra tiles to model factoring. In 7.2.3 Guided Practice, students shift to factoring algebraically without the algebra tiles. Throughout this lesson, students will engage with factoring trinomials using multiple methods and representations. Guide students from concrete to pictorial to abstract representations as understanding progresses.</td></tr>
<tr><td>Differentiate Instruction</td><td colspan="2">For 7.2.2 Exploration activity, consider providing students with actual algebra tile manipulatives, paper cutouts of the algebra tiles, or access to virtual algebra tiles for this lesson.

7.2.4 Your Turn! provides students with an independent practice opportunity factoring trinomials where the leading coefficient is one. Student performance on this activity can provide additional data on whether students may need additional support in future lessons.</td></tr>
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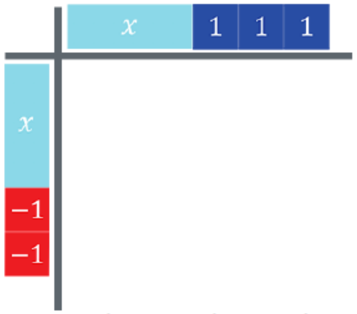
7.2.1 Warm-Up: Multiplying Binomials

***Component Overview**: Students will practice multiplying binomials using algebra tiles. For this activity, students work individually to recall multiplying binomials using algebra tiles from Unit 2 Lesson 6.*

7.2.1 Warm-Up: Multiplying Binomials

1. An algebra tile mat is shown with two binomial factors.



a. What are the two binomials represented?
$(x+3)$ *and* $(x-2)$

b. What is the product of the two binomials?
x^2+x-6

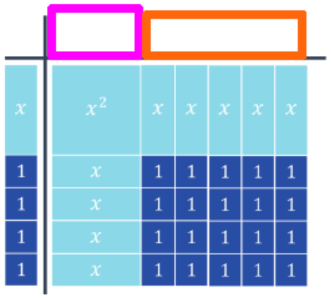
Consider spending time reviewing the process and answers to this component. Use this time to ensure that students are familiar with how to read and use algebra tiles. Connect the prior lesson's focus on factors and products to identifying where the factors and products are represented in this model.

7.2.2 Exploration: Model Factoring With Algebra Tiles

Component Overview: *Students will explore how to model factoring using algebra tiles. This activity is designed to be done with a partner.*

7.2.2 Exploration: Model Factoring with Algebra Tiles

1. An algebra tile mat is shown, where the product is the trinomial $x^2 + 9x + 20$ and one of its factors is $(x + 4)$.



Diego and Maria Jose both justified how they determined what belonged in the **pink** box in the following ways.

Diego	Maria Jose
I found $\frac{x^2}{x}$ because x^2 is the product of x and something else.	I asked myself what times x would give me x^2.

a. Discuss with your partner both justifications.
Answers will vary.

b. Use one of the methods to complete the binomial for the top bar of the algebra tile mat.

	Drawing	Algebraic Representation
Pink Box	x	x
Orange Box	1 1 1 1 1	5
Binomial Factor	$x + 5$	

This Exploration is scaffolded to allow students to engage with it with their partner rather than being guided through by the teacher. There will be an opportunity to summarize and synthesize what is discovered in the next component.

Observe and listen as students work with their partners. Make note of student ideas that you can ask them to share in the whole group during 7.2.3 Guided Practice.

Think-Pair-Share
Utilize this routine to give equitable opportunity for student thinking and learning. Students will be exploring in groups, so the Think-Pair-Share routine ensures that all students are engaged and not relying on one student to do the critical thinking.

Will Diego and Maria Jose get the same result based on what they have described? Why or why not?

7.2.2 Exploration: Model Factoring With Algebra Tiles (continued)

2. Discuss with your partner how to determine the two binomial factors when only given the product. Summarize your discussion.

Answers will vary. Find the dimensions of the rectangle.

x^2	x	x	x	x	x
x	1	1	1	1	1
x	1	1	1	1	1
x	1	1	1	1	1
x	1	1	1	1	1

The two binomials are the factors of the trinomial $x^2 + 9x + 20$. The process of finding the two binomials is called **factoring**.

3. The algebra tiles shown represent the trinomial $x^2 + 9x + 18$.

x^2	x	x	x
x	1	1	1
x	1	1	1
x	1	1	1
x	1	1	1
x	1	1	1
x	1	1	1

   a. Discuss with your partner what the similarities and differences are between the algebra tiles that represent $x^2 + 9x + 20$ and $x^2 + 9x + 18$. Summarize your discussion.

Similarities	*Answers will vary. Both of them are complete without any gaps in tiles. Both of them have one x^2 tile.*
Differences	*Answers will vary. The first model is wider horizontally and the second one is taller. One model has 18 unit tiles while the other has 20.*

   b. What are the two binomials that are factors of $x^2 + 9x + 18$?
   $(x + 3)$ and $(x + 6)$

Revisit this portion when reviewing the activity and highlight exemplar student work or responses. Consider charting up their responses and seeing which other students may agree or disagree with their rationale.

After students have shared out their thinking, provide a structured Think Aloud that helps students connect the product mat to the missing algebra tiles that represent the factors.

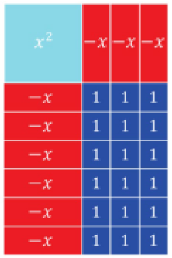
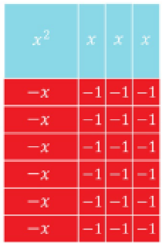
Connect this process and thinking to the narrative language about factoring on this page.

Encourage connections between concrete observation and abstract representation.

- *What do you notice about the algebra tiles?*
- *How might that be represented using algebra?*
- *What do you notice about the expression? How might that be represented or modeled using algebra tiles?*

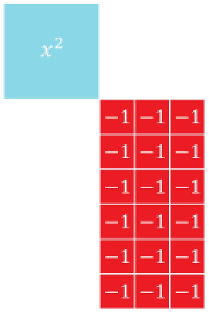
MA.K12.MTR.1.1: Participate in Effortful Learning
Consider having students share out their approaches. Students will benefit from hearing a variety of strategies from their peers.

4. The table shows the algebra tile representation for two trinomials. Work with your partner to determine the two binomial factors.

Trinomial	$x^2 - 9x + 18$	$x^2 - 3x - 18$
Algebra Tiles		
Binomial Factors	$(x - 3)$ *and* $(x - 6)$	$(x - 6)$ *and* $(x + 3)$

One common attribute of all of the algebra tile representations is that they are all rectangles.

5. Consider the following algebra tile representation.



a. Complete the algebra tile representation of a trinomial by drawing and labeling x or $-x$ tiles.
Three positive x tiles across the top and six negative x tiles along the side. Or: Three negative x tiles across the top and six positive x tiles along the side.

b. What is the trinomial that is represented by the completed drawing?
$x^2 - 3x - 18$, or alternatively $x^2 + 3x - 18$

c. What are the binomial factors of the trinomial?
$(x - 6)(x + 3)$ or alternatively, $(x + 6)(x - 3)$

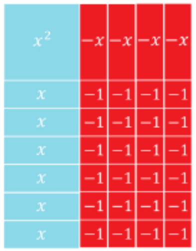
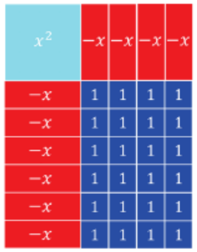
For question 4, it can be beneficial to have partner pairs compare their work with another partner pair or group.

Physical manipulatives can be very helpful with helping students think through question 5. If you don't have actual algebra tiles available, consider creating cutouts using card stock or different colored paper to represent the algebra tiles or providing access to virtual algebra tiles. For students who responded better to hands-on or visual stimuli, treating this like a puzzle can help them complete it and then connect it to the abstract concepts being explored.

Provide space for productive struggle. The Exploration builds intentionally for explicit connections between x-tiles and their values (positive or negative).

7.2.2 Exploration: Model Factoring With Algebra Tiles (continued)

6. The algebra tile representation of two different trinomials is shown.
   a. Complete the table.

Algebra Tiles	Trinomial	Factors
	$x^2 + 2x - 24$	$(x - 4)(x + 6)$
	$x^2 - 10x + 24$	$(x - 6)(x - 4)$

Each of the trinomials in the table has either twenty-four unit squares, $^{+}1$, or twenty-four negative unit squares, $^{-}1$. Also, each trinomial has some combination of four and six ^{+}x or ^{-}x tiles.

b. Discuss with your partner how the x-term of each trinomial relates to the four and six ^{+}x or ^{-}x tiles.
Answers will vary. The x term is the sum of the four and six ^{+}x or ^{-}x tiles.

c. Discuss with your partner how the constant term, 24 or -24, of each trinomial relates to the four and six ^{+}x or ^{-}x tiles.
Answers will vary. If the constant is positive, the x tiles would all need to be negative (or all positive). If the constant is negative, one set of the x tiles would need to be positive and the other negative.

Use informal observation up to this point to determine whether additional guidance may be needed prior to completing the Exploration. This could mean pulling a small group or teacher facilitation with the whole group to ensure understanding prior to releasing students to complete the remainder of the Exploration with their partners.

Use inquiry for synthesis when reviewing this activity.
- *What would it mean when the signs of the x-terms are different?*
- *How is the trinomial impacted when the signs of the x-terms are different?*

When reviewing the answer to part a, consider allowing different students to share the trinomial and each of the binomial factors. Allow them to engage in an agree/disagree protocol where they are polled on whether they agree with the answer or disagree.

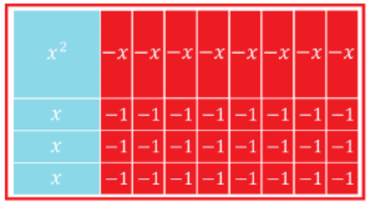
Leverage parts b and c from question 6 as a discussion point when reviewing as a whole group. The concepts developed here in these two questions are important for a deeper conceptual understanding.

7. Consider the trinomial $x^2 - 5x - 24$.

a. Work with your partner to determine how many ${}^{+}x$ or ${}^{-}x$ tiles are needed for $x^2 - 5x - 24$.

${}^{+}x$ **tiles**	${}^{-}x$ **tiles**
3	8

b. Draw a diagram of the algebra tiles that show the trinomial $x^2 - 5x - 2$



c. Use your algebra tiles diagram to choose the number of ${}^{+}x$ or ${}^{-}x$ tiles that match your diagram. Then, fill in each blank with the appropriate values.

	x	-8
x	x^2	○ $1x$ ○ $-1x$ ○ $6x$ ○ $-6x$ ○ $2x$ ○ $-2x$ ○ $8x$ ○ $-8x$ ○ $3x$ ○ $-3x$ ○ $12x$ ○ $-12x$ ○ $4x$ ○ $-4x$ ○ $24x$ ○ $-24x$
$+3$	○ $1x$ ○ $-1x$ ○ $6x$ ○ $-6x$ ○ $2x$ ○ $-2x$ ○ $8x$ ○ $-8x$ ○ $3x$ ○ $-3x$ ○ $12x$ ○ $-12x$ ○ $4x$ ○ $-4x$ ○ $24x$ ○ $-24x$	-24

d. Discuss with your partner how using the factors of -24, and the value of the coefficient of the b-term, -5, is helpful when factoring the trinomial $x^2 - 5x - 24$. Summarize your discussion.
Answers will vary. I can use the factors to determine which ones, when added together, will result in the middle term.

Students may struggle with question 7, since there are no algebra tiles presented already. Consider providing students with the tiles (or allowing them to use them if they already have them) to think through question 7 with their partners.

How does the diagram relate to the algebra-tile diagrams used up to this point?
Describe if switching where the number of positive and negative x-tiles are placed could give the same answer.

Listen as students discuss their thinking in response to part d to determine student thinking to highlight or misconceptions to address when transitioning to 7.2.3 Guided Practice.

7.2.3 Guided Practice: Factoring Trinomials

Component Overview: *Students will practice factoring trinomials where the leading coefficient is one.*

7.2.3 Guided Practice: Factoring Trinomials

Use observational data from the 7.2.2 Exploration to determine how much guidance is needed to bring together key ideas in this Guided Practice component. If necessary, consider modeling question 1 or pulling a small group as they work through the problems to address common misconceptions shown in the 7.2.2 Exploration.

For this activity, consider allowing students to complete using a Think-Pair-Share protocol. This will allow students to see what they know on their own and then engage in a collaborative conversation with their partner to clear up potential misconceptions.

1. Complete the table for $x^2 + 18x - 40$.

	x	$\underline{-2}$
x	x^2	○ $1x$ ○ $-1x$ ○ $8x$ ○ $-8x$ ○ $2x$ ○ $-2x$ ○ $10x$ ○ $-10x$ ○ $4x$ ○ $-4x$ ○ $20x$ ○ $-20x$ ○ $5x$ ○ $-5x$ ○ $40x$ ○ $-40x$
$\underline{+20}$	○ $1x$ ○ $-1x$ ○ $8x$ ○ $-8x$ ○ $2x$ ○ $-2x$ ○ $10x$ ○ $-10x$ ○ $4x$ ○ $-4x$ ○ $20x$ ○ $-20x$ ○ $5x$ ○ $-5x$ ○ $40x$ ○ $-40x$	-40

2. Complete the table for $x^2 + 12x + 32$.

	x	$\underline{+4}$
x	x^2	○ $1x$ ○ $-1x$ ○ $8x$ ○ $-8x$ ○ $2x$ ○ $-2x$ ○ $16x$ ○ $-16x$ ○ $4x$ ○ $-4x$ ○ $32x$ ○ $-32x$
$\underline{+8}$	○ $1x$ ○ $-1x$ ○ $8x$ ○ $-8x$ ○ $2x$ ○ $-2x$ ○ $16x$ ○ $-16x$ ○ $4x$ ○ $-4x$ ○ $32x$ ○ $-32x$	32

7.2.4 Your Turn!

Component Overview*: Students will independently practice factoring a trinomial where the leading coefficient is one.*

7.2.4 Your Turn!

1. Complete the table for $x^2 - 11x + 18$.

	x	$\underline{-9}$
x	x^2	○ $1x$ ○ $-1x$ ○ $6x$ ○ $-6x$ ○ $2x$ ○ $-2x$ ○ $9x$ ○ $-9x$ ○ $3x$ ○ $-3x$ ○ $18x$ ○ $-18x$
$\underline{-2}$	○ $1x$ ○ $-1x$ ○ $6x$ ○ $-6x$ ○ $2x$ ○ $-2x$ ○ $9x$ ○ $-9x$ ○ $3x$ ○ $-3x$ ○ $18x$ ○ $-18x$	18

Monitor students as they work independently through this component. Consider making note of which students may still be struggling, as this informal data can be used when you assign partners in the next lesson.

MA.K12.MTR.2.1: Multiple Representations
Encourage students to use algebraic phrases, algebra tiles, or other models and diagrams to represent either their process or their thinking.

7.2.5 Wrap-Up: Reflection

Component Overview: *Students will complete a self-reflection that has them determine how "in the zone" they feel about their understanding of factoring trinomials.*

7.2.5 Wrap-Up: Reflection

1. How "in the zone" do you feel right now about factoring trinomials?
   Answers will vary.

If students are unsure how to begin, consider using the learning target or benchmark from the lesson as sentence frames for students to construct their responses.